In[ ]:=
```
Quit[];
```

The Lorentzian quasi-Euclidean Kerr black hole (LEK)

In[1]:=
```
ClearAll["Global'*"];
list[y_] := y[[1]] == y[[2]];
rplc[y_] := y;
```

Metric definition

We start by defining the coordinates vector using the Doran coordinate system {t, r, θ, φ}

In[4]:=
```
x = {t, r, θ, φ}; n = 4;
```

and the function that implements the signature change ϵ

In[ ]:=
```
ϵ[r_] := ϵFunc[r];
```

We initialize the non-vanishing components of the LEK metric $g_{\mu\nu}$ in Doran coordinates

In[ ]:=
```
g11 = gtt = -ϵ[r] (1 - (2 M r) / (r^2 + a^2 Cos[θ]^2));
g22 = grr = (r^2 + a^2 Cos[θ]^2) / (r^2 + a^2);
g33 = gθθ = r^2 + a^2 Cos[θ]^2;
g44 = gφφ = (r^2 + a^2 + (2 M r a^2 Sin[θ]^2) / (r^2 + a^2 Cos[θ]^2)) Sin[θ]^2;
g14 = gtφ = -Sqrt[ϵ[r]] (2 M r a Sin[θ]^2) / (r^2 + a^2 Cos[θ]^2);
g12 = gtr = Sqrt[ϵ[r]] Sqrt[(2 M r) / (r^2 + a^2)];
g24 = grφ = -Sqrt[((2 M r) / (r^2 + a^2))] a Sin[θ]^2;
g23 = g34 = g13 = 0;
```

We now construct the metric tensor $g_{\mu\nu}$ and compute the inverse metric $g^{\mu\nu}$.
We also compute the determinant of the metric and verify the identity relation $g_{\mu\nu}\, g^{\mu\nu}$

```
In[ ]:= mt = FullSimplify[{{g11, g12, g13, g14},
          {g12, g22, g23, g24}, {g13, g23, g33, g34}, {g14, g24, g34, g44}}];
        it = FullSimplify[Inverse[mt]];
        mx = FullSimplify[
           Table[Simplify[Sum[it[[i, k]] * mt[[k, j]], {k, 1, n}]], {i, 1, n}, {j, 1, n}]];
        md = FullSimplify[Det[mt]];
        MatrixForm[mt]
        MatrixForm[it]
        MatrixForm[mx]
```

Out[ ]//MatrixForm=

$$\begin{pmatrix} \left(-1+\frac{2\,M\,r}{r^2+a^2\,\text{Cos}[\theta]^2}\right)\in\text{Func}[r] & \sqrt{2}\,\sqrt{\frac{M\,r}{a^2+r^2}}\,\sqrt{\in\text{Func}[r]} & 0 & -\frac{2\,a\,M\,r\,\text{Sin}[\theta]^2\,\sqrt{\in\text{Func}}}{r^2+a^2\,\text{Cos}[\theta]^2} \\ \sqrt{2}\,\sqrt{\frac{M\,r}{a^2+r^2}}\,\sqrt{\in\text{Func}[r]} & \frac{r^2+a^2\,\text{Cos}[\theta]^2}{a^2+r^2} & 0 & -\sqrt{2}\,a\,\sqrt{\frac{M\,r}{a^2+r^2}}\,\text{Sin}[\\ 0 & 0 & r^2+a^2\,\text{Cos}[\theta]^2 & 0 \\ -\frac{2\,a\,M\,r\,\text{Sin}[\theta]^2\,\sqrt{\in\text{Func}[r]}}{r^2+a^2\,\text{Cos}[\theta]^2} & -\sqrt{2}\,a\,\sqrt{\frac{M\,r}{a^2+r^2}}\,\text{Sin}[\theta]^2 & 0 & \text{Sin}[\theta]^2\left(a^2+r^2+\frac{2\,a^2\,M\,r}{r^2+a^2}\right. \end{pmatrix}$$

Out[ ]//MatrixForm=

$$\begin{pmatrix} -\frac{1}{\in\text{Func}[r]} & \frac{\sqrt{2}\,M\,r}{\sqrt{\frac{M\,r}{a^2+r^2}}\,\left(r^2+a^2\,\text{Cos}[\theta]^2\right)\,\sqrt{\in\text{Func}[r]}} & 0 & 0 \\ \frac{\sqrt{2}\,M\,r}{\sqrt{\frac{M\,r}{a^2+r^2}}\,\left(r^2+a^2\,\text{Cos}[\theta]^2\right)\,\sqrt{\in\text{Func}[r]}} & \frac{2\left(a^2+r\,(-2\,M+r)\right)}{a^2+2\,r^2+a^2\,\text{Cos}[2\,\theta]} & 0 & \frac{\sqrt{2}\,a\,\sqrt{\frac{M\,r}{a^2+r^2}}}{r^2+a^2\,\text{Cos}[\theta]^2} \\ 0 & 0 & \frac{1}{r^2+a^2\,\text{Cos}[\theta]^2} & 0 \\ 0 & \frac{\sqrt{2}\,a\,\sqrt{\frac{M\,r}{a^2+r^2}}}{r^2+a^2\,\text{Cos}[\theta]^2} & 0 & \frac{\text{Csc}[\theta]^2}{a^2+r^2} \end{pmatrix}$$

Out[ ]//MatrixForm=

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Curvature related quantities including explicit form of Ricci and regularized Kretschmann scalar

Christoffel symbols $\Gamma^{\alpha}{}_{\mu\nu}$

```
In[ ]:= chr = ParallelTable[FullSimplify[
            (1 / 2) Sum[(it[[i, s]]) (D[mt[[s, j]], x[[k]]] + D[mt[[s, k]], x[[j]]] - D[mt[[j, k]], x[[s]]]),
              {s, 1, n}]], {i, 1, n}, {j, 1, n}, {k, 1, n}];
```

Riemann curvature tensor $R^{\beta}{}_{\mu\nu\alpha}$

```
In[ ]:= rmn = ParallelTable[FullSimplify[D[chr[[i, j, l]], x[[k]]] - D[chr[[i, j, k]], x[[l]]] +
              Sum[chr[[s, j, l]] * chr[[i, k, s]] - chr[[s, j, k]] * chr[[i, l, s]], {s, 1, n}]],
           {i, 1, n}, {j, 1, n}, {k, 1, n}, {l, 1, n}];
```

Ricci tensor $R_{\mu\nu}$

```
In[ ]:= rcc = ParallelTable[
           FullSimplify[Sum[rmn[[i, j, i, l]], {i, 1, n}]], {j, 1, n}, {l, 1, n}];
```

Ricci scalar R

```
In[•]:= R = FullSimplify[Sum[it[[i, j]] × rcc[[i, j]], {i, 1, n}, {j, 1, n}]]
```

Out[•]=

$$\left(4\,M\,r^2\left(a^2+r^2\right)\in\text{Func}'[r]\left(-4\in\text{Func}[r]+M\in\text{Func}'[r]\right)+2\,M\left(a^2+2\,r^2+a^2\,\text{Cos}[2\,\theta]\right)\in\text{Func}'[r]\left(2\left(a^2+r^2\right)\in\text{Func}[r]-M\,r^2\in\text{Func}'[r]\right)+\left(a^2+2\,r^2+a^2\,\text{Cos}[2\,\theta]\right)^2\left(\in\text{Func}'[r]\left(4\,(M-r)\in\text{Func}[r]+\left(a^2+r\,(-2\,M+r)\right)\in\text{Func}'[r]\right)-2\left(a^2+r\,(-2\,M+r)\right)\in\text{Func}[r]\in\text{Func}''[r]\right)\right)\Big/\left(\left(a^2+2\,r^2+a^2\,\text{Cos}[2\,\theta]\right)^3\in\text{Func}[r]^2\right)$$

When sending epsilon derivatives to zero we recover the standard result for a Kerr black hole (R=0)

```
In[•]:= R /. {D[ϵ[r], r] → 0, D[D[ϵ[r], r] → 0, r]}
```

Out[•]=

```
0
```

Fully covariant Riemann tensor $R_{\mu\nu\alpha\beta}$

```
In[•]:= rcv = ParallelTable[FullSimplify[Sum[mt[[i, j]] * rmn[[j, k, l, m]], {j, 1, n}]],
    {i, 1, n}, {k, 1, n}, {l, 1, n}, {m, 1, n}];
```

Fully contravariant Riemann tensor $R^{\mu\nu\alpha\beta}$

```
In[•]:= rcn = ParallelTable[FullSimplify[
     Sum[it[[m, i]] * it[[h, j]] * it[[o, k]] * it[[p, l]] * rcv[[i, j, k, l]], {i, 1, n},
      {j, 1, n}, {k, 1, n}, {l, 1, n}]], {m, 1, n}, {h, 1, n}, {o, 1, n}, {p, 1, n}];
```

Kretschmann scalar $K = R_{\mu\nu\alpha\beta}R^{\mu\nu\alpha\beta}$ and regularized form with derivatives of ϵ nullified

```
In[•]:= krn = FullSimplify[
    Sum[rcv[[i, j, k, l]] * rcn[[i, j, k, l]], {i, 1, n}, {j, 1, n}, {k, 1, n}, {l, 1, n}]];
```

```
In[•]:= regkrn = krn /. {D[ϵ[r], r] → 0, D[D[ϵ[r], r], r] → 0} // FullSimplify
```

Out[•]=

$$-\left(\left(192\,M^2\left(a^2-2\,r^2+a^2\,\text{Cos}[2\,\theta]\right)\left(3\,a^4-56\,a^2\,r^2+8\,r^4+4\,a^2\left(a^2-14\,r^2\right)\text{Cos}[2\,\theta]+a^4\,\text{Cos}[4\,\theta]\right)\right)\Big/\left(a^2+2\,r^2+a^2\,\text{Cos}[2\,\theta]\right)^6\right)$$

Fully covariant Weyl tensor $C_{\mu\nu\alpha\beta}$

```
In[•]:= wcv = ParallelTable[
    FullSimplify[rcv[[i, j, k, l]] - 1/(n - 2) (mt[[i, k]] × rcc[[j, l]] - mt[[i, l]] × rcc[[j, k]] -
        mt[[j, k]] × rcc[[i, l]] + mt[[j, l]] × rcc[[i, k]]) +
      R/((n - 1) (n - 2)) (mt[[i, k]] × mt[[j, l]] - mt[[i, l]] × mt[[j, k]])],
    {i, 1, n}, {j, 1, n}, {k, 1, n}, {l, 1, n}];
```

Fully contravariant Weyl tensor $C^{\mu\nu\alpha\beta}$

```
In[•]:= wcn = ParallelTable[FullSimplify[
     Sum[it[[m, i]] * it[[h, j]] * it[[o, k]] * it[[p, l]] * wcv[[i, j, k, l]], {i, 1, n},
      {j, 1, n}, {k, 1, n}, {l, 1, n}]], {m, 1, n}, {h, 1, n}, {o, 1, n}, {p, 1, n}];
```

Weyl tensor $C^{\mu}{}_{\nu\alpha\beta}$

```
In[•]:= weyl = ParallelTable[FullSimplify[Sum[it[[i, s]] × wcv[[s, j, k, l]], {s, 1, n}]],
    {i, 1, n}, {j, 1, n}, {k, 1, n}, {l, 1, n}];
```

Weyl tensor norm C

In[ ]:=
```
cnorm = FullSimplify[
    Sum[wcv[[i, j, k, l]] * wcn[[i, j, k, l]], {i, 1, n}, {j, 1, n}, {k, 1, n}, {l, 1, n}]];
```

Final verification that Weyl tensor norm should be equal to Kretschmann scalar since we are in a vacuum spacetime

In[ ]:=
```
(cnorm - krn) /. {D[ϵ[r], r] → 0, D[D[ϵ[r], r], r] → 0} // FullSimplify
```

Out[ ]=
```
0
```

Numerical Integration of the Geodesic Equations

We start defining the smooth signature-change function

In[35]:=
```
rplus = M + √(M^2 - a^2);

ϵ[r_[τ_]] := (r[τ] - rplus)^(1/(2 λ+1)) / ((r[τ] - rplus)^2 + β)^(1/(2 (2 λ+1)));
```

We define the metric tensor and Christoffel symbols with proper time τ in Boyer-Lindquist coordinates

In[37]:=
```
g11 = gtt = -ϵ[r[τ]] (1 - 2 M r[τ] / (r[τ]^2 + a^2 Cos[θ[τ]]^2));

g22 = grr = (r[τ]^2 + a^2 Cos[θ[τ]]^2) / (r[τ]^2 + a^2 - 2 M r[τ]);

g33 = gθθ = r[τ]^2 + a^2 Cos[θ[τ]]^2;

g44 = gφφ = (r[τ]^2 + a^2 + 2 M r[τ] a^2 / (r[τ]^2 + a^2 Cos[θ[τ]]^2) Sin[θ[τ]]^2) Sin[θ[τ]]^2;

g12 = gtr = 0;

g14 = gtφ = -√(ϵ[r[τ]]) 2 a M r[τ] Sin[θ[τ]]^2 / (r[τ]^2 + a^2 Cos[θ[τ]]^2);

g13 = gtθ = 0;

g24 = grφ = 0;

g34 = gθφ = 0;

g23 = grθ = 0;
```

In[47]:=
```
xτ = {t[τ], r[τ], θ[τ], φ[τ]};
```

In[48]:=
```
mt = FullSimplify[{
     {g11, g12, g13, g14},
     {g12, g22, g23, g24},
     {g13, g23, g33, g34},
     {g14, g24, g34, g44}
    }];
```

In[49]:=
```
it = Simplify[Inverse[mt]];
```

In[50]:=
```
chr = ParallelTable[Simplify[(1 / 2)
      Sum[(it[[i, s]]) (D[mt[[s, j]], xτ[[k]]] + D[mt[[s, k]], xτ[[j]]] - D[mt[[j, k]], xτ[[s]]]),
       {s, 1, n}]], {i, 1, n}, {j, 1, n}, {k, 1, n}];
```

We define the four-velocity as first derivatives of the the position with respect to the affine parameter τ

In[51]:=
```
u = {D[xτ[[1]], τ], D[xτ[[2]], τ], D[xτ[[3]], τ], D[xτ[[4]], τ]};
uu = {ut[τ], ur[τ], uθ[τ], uφ[τ]};
eqs1 = uu[[1]] == u[[1]];
eqs2 = uu[[2]] == u[[2]];
eqs3 = uu[[3]] == u[[3]];
eqs4 = uu[[4]] == u[[4]];
```

We write down the geodesic equation $\frac{du^\mu}{d\tau} = -\Gamma^\mu_{\nu\gamma} u^\nu u^\gamma$

In[63]:=
```
geo =
  ParallelTable[-Sum[chr[[i, j, k]] × uu[[j]] × uu[[k]], {j, 1, n}, {k, 1, n}], {i, 1, n}];
equ = ParallelTable[{D[uu[[i]], τ] == rplc[geo[[i]]}, {i, 1, n}];
geodesic1 = equ[[1]][[1]];
geodesic2 = equ[[2]][[1]];
geodesic3 = equ[[3]][[1]];
geodesic4 = equ[[4]][[1]];
```

We set the parameters for the black hole and the LEK smooth function

In[69]:=
```
a = 0.99;
M = 1;
λ = 1;
β = 0.001;
```

Finally, we numerically integrate the geodesic equations. The following is configuration A from Table I in the Paper, but other cases can be obtained replacing the initial condition values

```
τf = 2000;
ni = NDSolve[{eqs1, eqs2, eqs3, eqs4, geodesic1, geodesic2, geodesic3,
     geodesic4, t[0] == 0, r[0] == 2.2, θ[0] == π / 2, φ[0] == π, ut[0] == 12.245,
     ur[0] == -0.001, uθ[0] == 2, uφ[0] == 3}, {t, r, θ, φ, ut, ur, uθ, uφ}, {τ, 0, τf}];
```

In[83]:=
```
traj = ParametricPlot3D[Evaluate[{Sqrt[r[τ]^2 + a^2] Sin[θ[τ]] Cos[φ[τ]],
        Sqrt[r[τ]^2 + a^2] Sin[θ[τ]] Sin[φ[τ]], r[τ] Cos[θ[τ]]} /. ni], {τ, 0, 100},
   PlotRange → {{-8, 8}, {-8, 8}, {-8, 8}}, PlotStyle → {Darker[Magenta], Thick}];
sfera = Graphics3D[{Opacity[0.9], Black, Sphere[{0, 0, 0}, rplus]}, Boxed → False];
Show[traj, sfera, Boxed → False, Axes → True,
 AxesLabel → {"x", "y", "z"}, ViewPoint → Front, Lighting → "Neutral"]
```

Out[ ]=

